

Case Studies

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[Practical]

Task 1: Create and Manage Files

Objective: Practice creating, listing, viewing, and deleting files

```
nikshitha@Nikshitha:~$ mkdir linux_task 1
nikshitha@Nikshitha:~$ cd linux_task1
nikshitha@Nikshitha:~/linux_task1$ touch file1.txt file2.txt file3.txt file4.txt file5.txt
nikshitha@Nikshitha:~/linux_task1$ ls
file1.txt file2.txt file3.txt file4.txt file5.txt
nikshitha@Nikshitha:~/linux_task1$ ls -l
total 4
-rw-r--r-- 1 nikshitha nikshitha 22 Aug  7 02:59 file1.txt
-rw-r--r-- 1 nikshitha nikshitha  0 Aug  7 02:59 file2.txt
-rw-r--r-- 1 nikshitha nikshitha  0 Aug  7 02:59 file3.txt
-rw-r--r-- 1 nikshitha nikshitha  0 Aug  7 02:59 file4.txt
-rw-r--r-- 1 nikshitha nikshitha  0 Aug  7 02:59 file5.txt
nikshitha@Nikshitha:~/linux_task1$ nano file1.txt
nikshitha@Nikshitha:~/linux_task1$ cat file1.txt
cat Hello, this is file1.
nikshitha@Nikshitha:~/linux_task1$ rm file5.txt
nikshitha@Nikshitha:~/linux_task1$ ls
file1.txt file2.txt file3.txt file4.txt
nikshitha@Nikshitha:~/linux_task1$ |
```

Task 2: Copy and Move Files

Objective: Practice copying and moving files and directories.

```

nikshitha@Nikshitha:~$ mkdir linux_task2
nikshitha@Nikshitha:~$ cd linux_task2
nikshitha@Nikshitha:~/linux_task2$ touch source.txt data.txt notes.txt
nikshitha@Nikshitha:~/linux_task2$ nano source.txt
nikshitha@Nikshitha:~/linux_task2$ mkdir backup
nikshitha@Nikshitha:~/linux_task2$ cp source.txt backup
nikshitha@Nikshitha:~/linux_task2$ ls backup
source.txt
nikshitha@Nikshitha:~/linux_task2$ cp data.txt backup/data_backup.txt
nikshitha@Nikshitha:~/linux_task2$ ls backup
data_backup.txt  source.txt
nikshitha@Nikshitha:~/linux_task2$ mv notes.txt backup/
nikshitha@Nikshitha:~/linux_task2$ ls
backup  data.txt  source.txt
nikshitha@Nikshitha:~/linux_task2$ ls backup
data_backup.txt  notes.txt  source.txt
nikshitha@Nikshitha:~/linux_task2$ mv source.txt original.txt
nikshitha@Nikshitha:~/linux_task2$ ls
backup  data.txt  original.txt
nikshitha@Nikshitha:~/linux_task2$ ls backup
data_backup.txt  notes.txt  source.txt
nikshitha@Nikshitha:~/linux_task2$ tree
.
├── backup
│   ├── data_backup.txt
│   ├── notes.txt
│   └── source.txt
├── data.txt
└── original.txt

2 directories, 5 files
nikshitha@Nikshitha:~/linux_task2$ |

```

Task 3: Working with head and tail

Objective: Practice viewing specific portions of a file

```

nikshitha@Nikshitha:~/linux_task2/linux_task3$ mkdir linux_task3
nikshitha@Nikshitha:~/linux_task2/linux_task3$ cd linux_task3
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3$ touch students.txt
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3$ nano students.txt
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3$ cat students.txt
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3$ nano students.txt
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3$ cat Students.txt
cat: Students.txt: No such file or directory
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3$ cat students.txt
Panduranga reddy
Padma
Nikshitha
Sathwika
Rishika
Rithwika
Archana
BhanuPriya
Harshika
Varsha
Srija
KeerthiPriya
Srikanth
Angela
JhansiPriya
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3$ head -5 students.txt
Panduranga reddy
Padma
Nikshitha
Sathwika
Rishika
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3$ head students.txt
Panduranga reddy
Padma
Nikshitha
Sathwika
Rishika
Rithwika
Archana
BhanuPriya
Harshika
Varsha
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3$ tail -5 students.txt
Srija
KeerthiPriya
Srikanth
Angela
JhansiPriya

```

```

nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3$ tail -5 students.txt
Srija
KeerthiPriya
Srikanth
Angela
JhansiPriya
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3$ tail students.txt
Rithwika
Archana
BhanuPriya
Harshika
Varsha
Srija
KeerthiPriya
Srikanth
Angela
JhansiPriya
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3$ |

```

Task 4: Organize Files

Objective: Organize files into different directories using mv

```
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3/organization$ mkdir organization
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3/organization$ cd organization
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3/organization/organization$ touch file1.txt file2.txt file3.txt image.jpg image2.jpg
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3/organization/organization$ ls
file1.txt file2.txt file3.txt image.jpg image2.jpg
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3/organization/organization$ mkdir documents
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3/organization/organization$ mkdir images
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3/organization/organization$ mv *.txt documents/
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3/organization/organization$ ls documents
file1.txt file2.txt file3.txt
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3/organization/organization$ mv *.jpg images/
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3/organization/organization$ ls images
image.jpg image2.jpg
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3/organization/organization$ ls
documents images
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3/organization/organization$ mv documents/file1.txt documents/notes.txt
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3/organization/organization$ ls documents
file2.txt file3.txt notes.txt
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3/organization/organization$ mv images/image1.jpg images/photo.jpg
mv: cannot stat 'images/image1.jpg': No such file or directory
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3/organization/organization$ ls image
ls: cannot access 'image': No such file or directory
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3/organization/organization$ ls images
image.jpg image2.jpg
nikshitha@Nikshitha:~/linux_task2/linux_task3/linux_task3/organization/organization$ tree
.
├── documents
│   ├── file2.txt
│   ├── file3.txt
│   └── notes.txt
└── images
    ├── image.jpg
    └── image2.jpg

3 directories, 5 files
```

Task 5: Directory Management

Objective: Practice creating nested directories and removing them


```

nikshitha@Nikshitha:~$ mkdir project
nikshitha@Nikshitha:~$ cd project
nikshitha@Nikshitha:~/project$ mkdir source/
nikshitha@Nikshitha:~/project$ mkdir source/c/
nikshitha@Nikshitha:~/project$ mkdir source/python/
nikshitha@Nikshitha:~/project$ mkdir docs/
nikshitha@Nikshitha:~/project$ mkdir backup/
nikshitha@Nikshitha:~/project$ touch source/c/file1.c
nikshitha@Nikshitha:~/project$ touch source/c/file2.c
nikshitha@Nikshitha:~/project$ touch source/c/file3.c
nikshitha@Nikshitha:~/project$ ls source/c/
file1.c file2.c file3.c
nikshitha@Nikshitha:~/project$ touch source/python/program1.py source/python/program2.py
touch: cannot touch 'source/python/program1.py': No such file or directory
nikshitha@Nikshitha:~/project$ touch source/python/program1.py source/python/program2.py
nikshitha@Nikshitha:~/project$ ls source/python/
program1.py program2.py
nikshitha@Nikshitha:~/project$ tree
.
├── backup
├── docs
├── source
│   ├── c
│   │   ├── file1.c
│   │   ├── file2.c
│   │   └── file3.c
│   └── python
│       ├── program1.py
│       └── program2.py
└──
6 directories, 5 files
nikshitha@Nikshitha:~/project$ rmdir backup
nikshitha@Nikshitha:~/project$ ls
docs source
nikshitha@Nikshitha:~/project$ mkdir backup
nikshitha@Nikshitha:~/project$ touch backup/data.txt
nikshitha@Nikshitha:~/project$ rmdir backup
rmdir: failed to remove 'backup': Directory not empty
nikshitha@Nikshitha:~/project$ rm backup/data.txt
nikshitha@Nikshitha:~/project$ ls backup
nikshitha@Nikshitha:~/project$ rmdir backup
nikshitha@Nikshitha:~/project$ ls
docs source
nikshitha@Nikshitha:~/project$ tree

```

```

docs source
nikshitha@Nikshitha:~/project$ tree
.
├── docs
├── source
│   ├── c
│   │   ├── file1.c
│   │   ├── file2.c
│   │   └── file3.c
│   └── python
│       ├── program1.py
│       └── program2.py
└──
5 directories, 5 files
nikshitha@Nikshitha:~/project$ |

```

Task 6: File Permissions Using chmod

Objective: Understand Linux file permissions.

```
nikshitha@Nikshitha:~/project$ mkdir linux_task4
nikshitha@Nikshitha:~/project$ cd linux_task4
nikshitha@Nikshitha:~/project/linux_task4$ touch public .txt private.txt script.sh
nikshitha@Nikshitha:~/project/linux_task4$ nano public .txt
nikshitha@Nikshitha:~/project/linux_task4$ cat public.txt
cat: public.txt: No such file or directory
nikshitha@Nikshitha:~/project/linux_task4$ nano public.txt
nikshitha@Nikshitha:~/project/linux_task4$ cat public.txt
This is a public file.
nikshitha@Nikshitha:~/project/linux_task4$ nano private.txt
nikshitha@Nikshitha:~/project/linux_task4$ nano script.sh
nikshitha@Nikshitha:~/project/linux_task4$ ls -l
total 16
-rw-r--r-- 1 nikshitha nikshitha 26 Aug  7 03:49 private.txt
-rw-r--r-- 1 nikshitha nikshitha 24 Aug  7 03:48 public
-rw-r--r-- 1 nikshitha nikshitha 23 Aug  7 03:48 public.txt
-rw-r--r-- 1 nikshitha nikshitha  1 Aug  7 03:50 script.sh
nikshitha@Nikshitha:~/project/linux_task4$ chmod 600 private.txt
nikshitha@Nikshitha:~/project/linux_task4$ chmod 644 public.txt
nikshitha@Nikshitha:~/project/linux_task4$ chmod +x script.sh
nikshitha@Nikshitha:~/project/linux_task4$ ls -l
total 16
-rw----- 1 nikshitha nikshitha 26 Aug  7 03:49 private.txt
-rw-r--r-- 1 nikshitha nikshitha 24 Aug  7 03:48 public
-rw-r--r-- 1 nikshitha nikshitha 23 Aug  7 03:48 public.txt
-rwxr-xr-x 1 nikshitha nikshitha  1 Aug  7 03:50 script.sh
nikshitha@Nikshitha:~/project/linux_task4$ ./script.sh
nikshitha@Nikshitha:~/project/linux_task4$ ls -l
total 16
-rw----- 1 nikshitha nikshitha 26 Aug  7 03:49 private.txt
-rw-r--r-- 1 nikshitha nikshitha 24 Aug  7 03:48 public
-rw-r--r-- 1 nikshitha nikshitha 23 Aug  7 03:48 public.txt
-rwxr-xr-x 1 nikshitha nikshitha  1 Aug  7 03:50 script.sh
nikshitha@Nikshitha:~/project/linux_task4$ chmod -x script.sh
nikshitha@Nikshitha:~/project/linux_task4$ ls -l
total 16
-rw----- 1 nikshitha nikshitha 26 Aug  7 03:49 private.txt
-rw-r--r-- 1 nikshitha nikshitha 24 Aug  7 03:48 public
-rw-r--r-- 1 nikshitha nikshitha 23 Aug  7 03:48 public.txt
-rw-r--r-- 1 nikshitha nikshitha  1 Aug  7 03:50 script.sh
nikshitha@Nikshitha:~/project/linux_task4$ |
```

Task 7: File Permission Challenge

Objective: Understand how permissions affect files and directories

```

nikshitha@Nikshitha:~$ mkdir Permission_lab
nikshitha@Nikshitha:~$ cd Permission_lab
nikshitha@Nikshitha:~/Permission_lab$ touch test.txt
nikshitha@Nikshitha:~/Permission_lab$ nano test.txt
nikshitha@Nikshitha:~/Permission_lab$ ls -l
total 4
-rw-r--r-- 1 nikshitha nikshitha 23 Aug  7 04:02 test.txt
nikshitha@Nikshitha:~/Permission_lab$ chmod u-w test.txt
nikshitha@Nikshitha:~/Permission_lab$ ls -l
total 4
-r--r--r-- 1 nikshitha nikshitha 23 Aug  7 04:02 test.txt
nikshitha@Nikshitha:~/Permission_lab$ nano test.txt
nikshitha@Nikshitha:~/Permission_lab$ chmod u+w test.txt
nikshitha@Nikshitha:~/Permission_lab$ ls -l
total 4
-rw-r--r-- 1 nikshitha nikshitha 23 Aug  7 04:02 test.txt
nikshitha@Nikshitha:~/Permission_lab$ ./test.txt
-bash: ./test.txt: Permission denied
nikshitha@Nikshitha:~/Permission_lab$ |

```

Task 8: Backup and Restore

Objective: Create backups and restore files using cp.

```

nikshitha@Nikshitha:~$ mkdir linux_task6
nikshitha@Nikshitha:~$ cd linux_task6
nikshitha@Nikshitha:~/linux_task6$ touch important.txt
nikshitha@Nikshitha:~/linux_task6$ nano important.txt
nikshitha@Nikshitha:~/linux_task6$ mkdir backup
nikshitha@Nikshitha:~/linux_task6$ cp important.txt backup/
nikshitha@Nikshitha:~/linux_task6$ tree
.
├── backup
│   └── important.txt
└── important.txt

2 directories, 2 files
nikshitha@Nikshitha:~/linux_task6$ mv backup/important.txt backup/important_backup.txt
nikshitha@Nikshitha:~/linux_task6$ tree
.
├── backup
│   ├── important_backup.txt
│   └── important.txt
└── important.txt

2 directories, 2 files
nikshitha@Nikshitha:~/linux_task6$ rm important.txt
nikshitha@Nikshitha:~/linux_task6$ tree
.
├── backup
│   └── important_backup.txt
└── important_backup.txt

2 directories, 1 file
nikshitha@Nikshitha:~/linux_task6$ ls
backup
nikshitha@Nikshitha:~/linux_task6$ cp backup/important_backup.txt
cp: missing destination file operand after 'backup/important_backup.txt'
Try 'cp --help' for more information.
nikshitha@Nikshitha:~/linux_task6$ cp backup/important_backup.txt .
nikshitha@Nikshitha:~/linux_task6$ ls
backup  important_backup.txt
nikshitha@Nikshitha:~/linux_task6$ tree
.
├── backup
│   └── important_backup.txt
└── important_backup.txt

2 directories, 2 files
nikshitha@Nikshitha:~/linux_task6$ mv important_backup.txt important_restored.txt
nikshitha@Nikshitha:~/linux_task6$ cat important_restored.txt
This is a important file.
nikshitha@Nikshitha:~/linux_task6$ |

```


Task 9: Complete Linux File Management Challenge

Objective: Use almost all the commands together.

```
nikshitha@Nikshitha:~$ mkdir linux_project
nikshitha@Nikshitha:~$ cd linux_project
nikshitha@Nikshitha:~/linux_project$ mkdir source backup docs
nikshitha@Nikshitha:~/linux_project$ touch source/main.c source/helper.c source/README.t
nikshitha@Nikshitha:~/linux_project$ ls source/
README.txt  helper.c  main.c
nikshitha@Nikshitha:~/linux_project$ nano source/README.txt
nikshitha@Nikshitha:~/linux_project$ head -5 source/README.txt
Operating System Lab
Linux Commands
File Management
Directory Management
Permissions
nikshitha@Nikshitha:~/linux_project$ tail -5 source/README.txt
Operating System Lab
Linux Commands
File Management
Directory Management
Permissions
nikshitha@Nikshitha:~/linux_project$ tree
.
├── backup
├── docs
└── source
    ├── README.txt
    ├── helper.c
    └── main.c

4 directories, 3 files
nikshitha@Nikshitha:~/linux_project$ cp source/main.c backup/
nikshitha@Nikshitha:~/linux_project$ ls backup/
main.c
nikshitha@Nikshitha:~/linux_project$ ls
backup  docs  source
nikshitha@Nikshitha:~/linux_project$ cp source/README.txt docs/
nikshitha@Nikshitha:~/linux_project$ ls docs
README.txt
nikshitha@Nikshitha:~/linux_project$ mv source/helper.c source/function.c
nikshitha@Nikshitha:~/linux_project$ ls source
README.txt  function.c  main.c
nikshitha@Nikshitha:~/linux_project$ chmod 644 source/README.txt
nikshitha@Nikshitha:~/linux_project$ rm backup/main.c
nikshitha@Nikshitha:~/linux_project$ ls backup
nikshitha@Nikshitha:~/linux_project$ rmdir backup/
nikshitha@Nikshitha:~/linux_project$ ls
docs  source
nikshitha@Nikshitha:~/linux_project$ tree
```

```
docs  source
nikshitha@Nikshitha:~/linux_project$ tree
.
├── docs
│   └── README.txt
└── source
    ├── README.txt
    ├── function.c
    └── main.c

3 directories, 4 files
nikshitha@Nikshitha:~/linux_project$ |
```

Task 10: Mini Linux Administration Challenge

Objective: Complete a real-world-style file

```
nikshitha@Nikshitha:~$ mkdir student_portal
nikshitha@Nikshitha:~$ cd student_portal
nikshitha@Nikshitha:~/student_portal$ mkdir src docs backup logs config
nikshitha@Nikshitha:~/student_portal$ ls
backup  config  docs  logs  src
nikshitha@Nikshitha:~/student_portal$ touch src/main.c src/student.c docs/README.txt config/config.txt logs/system.log
nikshitha@Nikshitha:~/student_portal$ tree
.
├── backup
├── config
│   └── config.txt
├── docs
│   └── README.txt
├── logs
│   └── system.log
└── src
    ├── main.c
    └── student.c

6 directories, 5 files
nikshitha@Nikshitha:~/student_portal$ nano logs/system.log
nikshitha@Nikshitha:~/student_portal$ head -5 logs/system.log
System Started
User Login
Configuration Loaded
Database Connected
Service Started
nikshitha@Nikshitha:~/student_portal$ nano logs/system.log
nikshitha@Nikshitha:~/student_portal$ head -5 logs/system.log

System Started
User Login
Configuration Loaded

Database Connected
Service Started
nikshitha@Nikshitha:~/student_portal$ nano logs/system.log
nikshitha@Nikshitha:~/student_portal$ nano logs/system.log
nikshitha@Nikshitha:~/student_portal$ tail -5 logs/system.log
Memory Checked
Permission Verified
Files Synced
System Healthy
System Shutdown
nikshitha@Nikshitha:~/student_portal$ cp docs/README.txt backup/
nikshitha@Nikshitha:~/student_portal$ ls backup
```

```
nikshitha@Nikshitha:~/student_portal$ cp docs/README.txt backup/
nikshitha@Nikshitha:~/student_portal$ ls backup
README.txt
nikshitha@Nikshitha:~/student_portal$ mv backup/README.txt backup/README_backup.txt
nikshitha@Nikshitha:~/student_portal$ ls backup
README_backup.txt
nikshitha@Nikshitha:~/student_portal$ chmod 600 config/config.txt
nikshitha@Nikshitha:~/student_portal$ ls -l config
total 0
-rw----- 1 nikshitha nikshitha 0 Aug  7 06:24 config.txt
nikshitha@Nikshitha:~/student_portal$ chmod 644 docs/README.txt
nikshitha@Nikshitha:~/student_portal$ ls -l docs
total 0
-rw-r--r-- 1 nikshitha nikshitha 0 Aug  7 06:24 README.txt
nikshitha@Nikshitha:~/student_portal$ rm backup/README_backup.txt
nikshitha@Nikshitha:~/student_portal$ ls backup
nikshitha@Nikshitha:~/student_portal$ rmdir backup
nikshitha@Nikshitha:~/student_portal$ ls
config docs logs src
nikshitha@Nikshitha:~/student_portal$ tree
.
├── config
│   └── config.txt
├── docs
│   └── README.txt
├── logs
│   └── system.log
└── src
    ├── main.c
    └── student.c

5 directories, 5 files
nikshitha@Nikshitha:~/student_portal$ w|
```